

Robotics Teaching with Interactive Dashboards: A Visual Machine Learning Methodology for Engineering

Author names omitted for double-blind review – Latin.Science 2026

Abstract—This study evaluates an interactive, simulation-based educational platform designed for robotics instruction and reinforcement learning in higher education. By combining visual simulation with direct parameter manipulation, the system fosters student autonomy in solving complex engineering challenges, such as robotic navigation and obstacle negotiation. As students engage with the software, interaction data is continuously collected and organized into a structured repository, providing the baseline parameters required to train reinforcement learning algorithms for physical robot deployment. The lightweight computational architecture ensures hardware accessibility and cross-platform compatibility, enabling scalable adoption in higher education environments. Furthermore, the platform features a dynamic dashboard that allows real-time modifications of robotic morphologies, ranging from quadrupedal platforms such as the Go2 to humanoid models, alongside environmental parameters including terrain inclination and surface friction. Future enhancements involve integrating an open-source programming interface supporting Python to facilitate custom scenario development. Overall, the findings demonstrate that the platform effectively enhances student comprehension of core robotics concepts and reinforcement learning methodologies.

Keywords—Robotics Education, Reinforcement Learning, Physics Simulation, Higher Education, Software Accessibility.

STATEMENT ON THE USE OF ARTIFICIAL INTELLIGENCE Generative Artificial Intelligence tools were used in this work to assist with drafting and editing the text and with translating it into English. All AI-generated content was critically reviewed and validated by the authors, who assume full responsibility for the originality and accuracy of the final text, in accordance with CNPq Ordinance No. 2.664/2026.

1 Introduction

Recent technological evolution has significantly transformed global production processes and driven the expansion of robotics across various industrial and academic sectors. This highly interdisciplinary field integrates concepts from mechanical engineering, computing, and intelligent systems for the automation and execution of complex tasks. Although

the study or mastery of these technologies is fundamental in contemporary engineering education, teaching robotics faces significant methodological challenges in the university environment. The main barrier lies in the practical presentation of abstract concepts, such as machine learning, dynamic control, and reinforcement learning, which often remain restricted to theoretical or briefly demonstrative mathematical formulations. Through the system developed in this work, it becomes possible to execute real-time kinematic and locomotion simulations, as well as to explore interactive resources that traditional teaching based on theoretical lectures cannot easily provide.

The primary pillar of this work stems from the constant challenge faced by educators in seeking innovation in their pedagogical practices in higher education. Robotics-mediated education has consolidated itself as an efficient strategy, capable of significantly contributing to the development of technical skills, logical reasoning, and complex problem-solving [1]. As a proposal to mitigate accessibility and practical feasibility limitations, this paper presents the development and application of an interactive platform integrated with the MuJoCo physics simulation environment. The tool consists of a web dashboard designed primarily for students and professionals with little to no prior knowledge on the subject. Through this simplified interface, the user can directly observe how the quadruped robot's actions, including joint angle actuation and horizontal displacement across terrain, relate to input variables and parameters. The central objective is to make the learning process more practical, visual, and accessible, eliminating reliance on highly complex programming interfaces and the need for expensive physical hardware with delicate maintenance, effectively using the virtual environment as an intuitive tool.

In intelligent systems, machine learning focuses on developing algorithms and computational models capable of acquiring knowledge and inferring patterns autonomously from accumulated experiences [2]. As one of the main paradigms in this broad area, reinforcement learning operates based on a dynamic system of rewards and penalties, in which the agent being tested iteratively adjusts its decision-making based on the outcomes obtained from each action executed in the environment. However, the practice of assimilating produced data, state matrices, and reward functions usually requires considerable effort and advanced knowledge from students. The use of the interactive dashboard directly facilitates the understanding of these theoretical concepts by

translating mathematical ideas into immediate visual feedback, directly contributing to the enhancement of pedagogical approaches in engineering education.

Finally, this work aims to demonstrate how the platform can support higher education and robotics by combining interactive data visualization, execution in lightweight computing environments, and a practical introduction to the fundamentals of Machine Learning. This pedagogical approach generates a positive impact on Engineering education, providing practical support that empowers future professionals to understand autonomous agent dynamics and develop innovative, automated solutions for the continuous advancement of the technology sector.

2 Conceptualization

The conceptual framework explaining the development and validation of the proposed dashboard will be detailed. Initially, the criteria for selecting the computing environment—which in this case was MuJoCo—and the choice of the quadruped robot model are presented, correlating their physical capabilities with real-time simulation feasibility. Next, introductory concepts of the simulator are established, highlighting its accuracy as a virtual environment, alongside the theoretical foundations of Reinforcement Learning, addressing the decision-making process formulation and reward propagation. Additionally, the methodology employed to monitor and validate learning progress is outlined, demonstrating how the lightweight and intuitive interface parameters enable practical analysis and validation of robotics concepts.

2.1 The MuJoCo Environment

MuJoCo is an advanced physics simulation engine designed for simulating complex dynamic systems with high precision and performance, particularly in robotics, mechanics, and machine learning. Although the platform can act as a complete training environment, its interface is not very intuitive for beginners, which can cause initial resistance. Thus, while MuJoCo constitutes the base framework for applying Reinforcement Learning, one of the central proposals of this project is to present these concepts through a more user-friendly and accessible interface.

2.2 Reinforcement Learning and Quadruped Robots

Reinforcement Learning relies on autonomous interaction between an agent and a given environment. The agent performs an action and receives feedback in the form of a reward or penalty. The term "reinforcement" connects to this adjustment mechanism: positive rewards signal the success of a decision—for example, if the robot steps up a staircase in a manner consistent with the defined goal, the system provides a bonus for achieving a portion of the final result, encouraging its repetition—whereas penalties (negative rewards) indicate failure, discouraging the behavior, such as a quadruped robot misstepping and falling. The latter represents a setback toward the goal of climbing stairs, but if it successfully

supports itself on a step, we positively reinforce it. This interaction cycle is summarized in the following diagram.

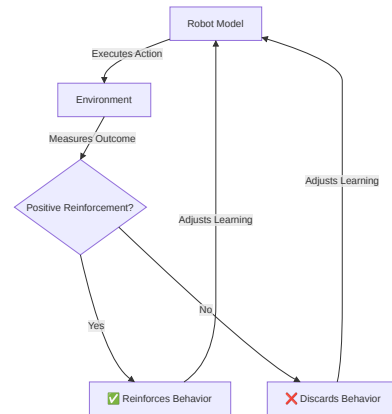


Figure 1: Interaction and evaluation cycle in Reinforcement Learning.

Considering the operation of the base platform and the principles of reinforcement learning, the Unitree Go2 quadruped robot was adopted as the reference model for training. This choice is justified by the need to maintain an introductory pedagogical approach, prioritizing initial simplicity in the process. This model was chosen because its foundation is simpler: it has four legs and twelve joints in total (three joints per leg), corresponding to the degrees of freedom for the hip, thigh, calf, and knee.

2.3 Dashboards and Interactive Interfaces

The adoption of the dashboard as a pedagogical tool is justified by its ability to integrate the display of simulation data with active control by the student. The interface allows monitoring and manipulating variables such as robot movement, success and failure rates of training, and terrain characteristics. Consequently, students develop the skill to interpret the provided data matrix, understanding how to filter and apply this information according to the established learning goals.

2.4 Applied Pedagogy in Robotics

Consolidating learning in complex domains requires immersive and structured pedagogical practices. According to César Coll and Elena Martín, knowledge assimilation is directly linked to the teacher's role and methodologies, moving beyond the view that assessment depends solely on student effort [3]. In Brazilian higher education within technology fields, providing interactive tools is essential to increase efficiency and engagement in the educational process. However, the feasibility of this strategy is tied to the functional efficiency of the platform and the effectiveness of its actual learning gains for students.

3 Graphical Interface of the Dashboard and MuJoCo

To introduce the features and structure developed to complement MuJoCo, specifically exploring the learning component, we will demonstrate the interface and its tools. This includes the partial practical operation of the MuJoCo platform.



Figure 2: Main Dashboard Interface

The platform features joint angle and movement indicators, as well as initial poses and parameters for initializing the simulation, allowing the execution of basic routines such as forward and backward movement.

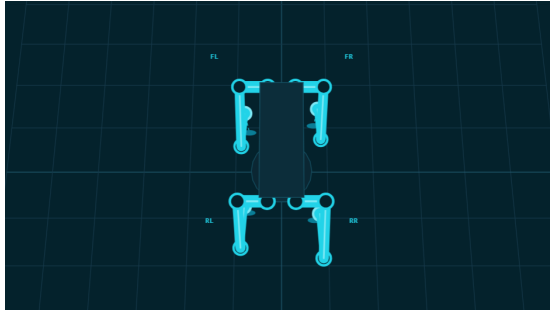


Figure 3: Go2 Top-View Simulation

Figure 3 illustrates the integration of the model with the tool, where all instructions configured on the dashboard are reflected in MuJoCo. Next, the simulation environment applies Reinforcement Learning algorithms and transmits the gathered data and metrics back to the graphical interface.



Figure 4: Real-time information matrices during simulation

Figure 4 presents the matrices responsible for capturing the movement variables of the quadruped robot during simulation. This data is processed in real-time, providing the

necessary feedback for RL (Reinforcement Learning) training.



Figure 5: MuJoCo

Figure 5 shows the robotic model in its native simulation environment in MuJoCo.

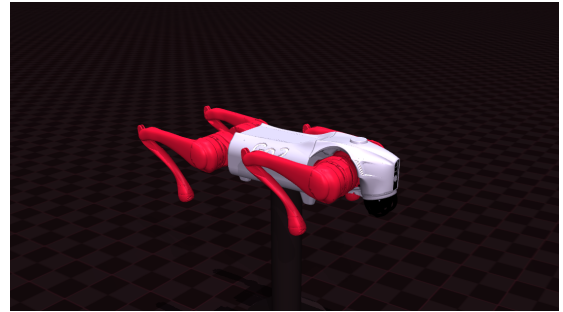


Figure 6: Robotic dog crouching in MuJoCo

Figure 6 demonstrates the Unitree robot crouching as a mirroring of the dashboard shown in Figure 2, proving that both are equally interactive.

4 Methodology

The methodology employed in developing the platform and its integrated operation is structured into fundamental steps to ensure a concise and demonstrative foundation. First, the system architecture is detailed, including the choice of programming languages, frameworks, and the rationale for the technological decisions made. Next, the computational integration between the developed interface and the MuJoCo simulator is addressed, explaining the dynamic bidirectional interaction and the benefits of this architecture. Finally, the pedagogical potential of the platform is discussed, demonstrating the methodology's suitability for introductory Machine Learning education, alongside a qualitative analysis of the tool's educational impact. In terms of education, the system was designed to be intuitive, lightweight, and easy to use, presenting a receptive interface that enables practical experimentation. The platform offers multiple modes for robot training, starting with the execution of basic walking cycles that foster the concept of reinforcement learning. This dynamic encourages critical analysis from the user and provides a clear visual of how varying parameters and numbers directly impacts the model's behavior, integrating theory with practice.

4.1 Platform Architecture

To establish integration and direct control over the MuJoCo simulation environment, Python was selected for system development due to its versatility, ease of use, and web integration capabilities.

For visual rendering and physical calculations of the robot, the code uses the aforementioned MuJoCo, a specialized 3D simulation tool. With it, the program can render the quadruped robot on screen, calculate gravity, leg movements, floor impacts, and other commands issued by our dashboard. The mathematics behind the robot's angles and positions are computed using the NumPy library, widely used for numerical processing.

To communicate with the control screen, the script utilizes the local network via the HTTP protocol. It constantly checks whether a button on the screen has been pressed to respond instantly. To ensure the robot moves without freezing the screen, the code employs a technique called multiprocessing. This allows the computer to handle two tasks simultaneously: one part of the program is dedicated solely to communicating with the web screen over the internet, while the other part manages the 3D rendering and robot physics at 100 frames per second. Finally, the code is intelligent enough to automatically download the robot's 3D files and configurations from the internet if they are not yet present on the computer, while also accepting simple startup commands via the terminal. In practice, all user-requested actions, including jumps, dodges, and manual or autonomous movement commands, are reflected immediately and seamlessly in the MuJoCo simulator.

4.2 Dashboard Integration with MuJoCo

The connection between the web application and MuJoCo occurs through a continuous real-time data stream, operating in a constant update cycle synchronized with the simulation. This stream is facilitated by an asynchronous communication structure that captures network-sent instructions and processes them without causing visual rendering bottlenecks. As soon as data arrives, the system instantly recalculates force vectors, mechanical limits, and environment collision geometries, translating every user interaction into an immediate physical response. Beyond reflecting joint and posture changes on the robot with low latency, this architecture dynamically reconfigures scene elements and recalculates visual framing trajectories, keeping the camera perfectly aligned with 3D spatial movements.

4.3 Disciplined Didactics of the Platform

The use of digital tools, such as programming software, has proven essential for fostering a deeper understanding of concepts through experimentation and practice, providing meaningful learning [5][6]. In this sense, the innovation of the created system lies in democratizing robotics education by connecting an intuitive interface to a moderately complex simulator via an optimized, lightweight architecture. By abstracting mathematical concepts and algorithms into simple parameters, the platform allows students to explore quadruped robot behavior, practically understanding how joint varia-

tions and new locomotion methods propagate in the model's response. This approach eliminates the need for expensive computing infrastructures or costly hardware, turning learning into an experimental process where students themselves configure scenarios, test hypotheses, and visually observe the real-time impact of each adjustment.

5 System Effectiveness Evaluation

As discussed in previous sections, much has been said regarding the intuitiveness, effectiveness, lightweight integration, and pedagogical potential of the platform. To validate these characteristics, its functionality will be demonstrated practically, starting with the execution of a basic walking cycle. In this setup, the system acts as the motion pattern generator, with results replicated in the MuJoCo environment. The demonstration is divided into two stages: first, analyzing the joint response of a single leg in a specific frame.



Figure 7: Robotic dog taking continuous steps forward

It can be observed that during simulation, results become even clearer when graphs highlight movement precisely, matching the output in MuJoCo, as shown in Figure 8.



Figure 8: Robotic dog performing the same task in MuJoCo



Figure 9: Analytical result of the moving leg

In Figure 9, the angular variation graph reflecting front joint movement (Q3) shows the statistical behavior of the motion under limited amplitude.

In a complementary simulation, quadruped movement supported by all four legs is demonstrated, validating motor coordination and sequential alternation of joint commands during the walking cycle.



Figure 10: *Walking demonstration on the Dashboard*



Figure 11: *Walking demonstration on the Dashboard*

6 Discussion

Pedagogical innovation in higher education is a complex and multifaceted phenomenon that can take very diverse forms [7]; thus, an effective methodology is one where both student and educator resonate with the content. Teaching robotics through direct practice and simulation offers high learning potential. System interactivity and guided manipulation give students autonomy in the learning process, empowering them to solve practical engineering challenges, such as navigation control and obstacle negotiation. As students run successive tests in the application, the system stores interaction history and generates a structured database. This repository of experiences can be used to train Reinforcement Learning algorithms, providing resources for subsequent parameter transfer to physical robots—that is, using the virtual environment to accumulate data and transfer it. The software architecture, characterized by low computational resource consumption and high accessibility, enables execution on mid/low-spec hardware. This feature facilitates method replication in larger student cohorts and across varying technological infrastructures. For future work, training effectiveness should be validated across larger student samples and expanded to different robot morphologies, including humanoid models and advanced quadruped platforms such as the Go2 model. The platform supports dynamic reconfiguration of these models through the control panel, allowing

updates to joint constraints, range-of-motion limits, and environmental variables such as friction coefficient and terrain slope. Introducing an open-source module supporting widely used programming languages like Python would also be a valuable addition. This functionality will allow users to create custom routines and simulate complex scenarios. The obtained results demonstrate that the tool contributes to consolidating fundamental concepts in robotics and reinforcement learning.

7 Conclusion

Considering that youth interest in robotics makes it a viable approach for introductory programming education [7], the developed system goes beyond a conventional simulation tool. Serving as a practical methodology for visualizing and understanding concepts related to machine learning and quadruped robot motion mechanics, it can also serve as an incentive for education in other fields. Furthermore, the platform enables mathematical engagement, such as interpreting analytical graphs and matrices, broadening its potential as an educational resource.

This paper presented an interactive platform developed to make understanding the processes involved in machine learning model training and robotic dog movement more accessible. The obtained results indicate that the platform features an intuitive interface, easy to use and access, enabling application regardless of the user's specific hardware specs.

The integration among robot movement, data collection and emission, model training, and the platform itself proved functional in providing an introduction to machine learning and reinforcement learning concepts. Thus, the system shows potential for use by both beginners and professionals with limited experience in the field, facilitating visual and interactive comprehension of necessary model training steps.

Additionally, the platform holds potential for future applications related to collaborative teaching and practical experimentation. As a continuation of this work, data generated by the platform could be used to train a physical robotic dog, allowing the trained model to interact directly with the environment and use reinforcement learning techniques to refine its movements and behaviors. This possibility highlights the potential of the proposed integration between simulation, data collection, training, and robotic control, making the platform a promising tool for robotics and machine learning education. However, it is important to highlight certain limitations of this work. The absence of dynamic testing under conditions closer to real-world applications and the limited amount of experimental data do not yet allow conclusive claims regarding consistent system performance across varied scenarios. Future continuations will expand dynamic testing and utilize platform-generated data to train a robotic dog in the simulated environment, allowing the model to interact directly with the environment and use reinforcement learning techniques to improve its movements and behaviors.

References

- [1] C. B. Noronha and A. J. C. da Silva, “Educational robotics in Brazil: a systematic literature review,” *Revista FT*, 2025.
- [2] M. C. Monard and J. A. Baranauskas, “Concepts on Machine Learning,” in *Intelligent Systems for Engineering and Medicine*, S. O. Rezende, Ed. Rio de Janeiro: Manole, 2003, ch. 4, pp. 89–114.
- [3] C. Coll et al., *Constructivism in the Classroom*, 6th ed. São Paulo: Ática, 1999.
- [4] T. F. B. Nunes, C. C. Viana, and L. A. F. C. Viana, “Perspectives of robotics as a pedagogical resource applied to Education 4.0: a bibliometric analysis on educational robotics,” *Research, Society and Development*, vol. 10, no. 4, p. e6310413889, 2021. DOI: 10.33448/rsd-v10i4.13889.
- [5] S. M. O. Riboldi and J. T. Reichert, “The use of digital tools in teaching,” in *Proceedings of the Congress on Technology in Education*, 2020.
- [6] F. S. Sápiras, R. D. Vecchia, and M. V. Maltempi, “The use of technologies and software in the teaching-learning process,” *Educação Matemática Pesquisa*, vol. 17, no. 3, pp. 520–535, 2015.
- [7] T. Almeida and J. F. Netto, “Pedagogical Robotics Applied to Programming Teaching: A Systematic Literature Review,” in *Proceedings of the 26th Brazilian Symposium on Computers in Education (SBIE)*, Maceió, Brazil, 2015, pp. 597–606. DOI: 10.5753/cbie.sbie.2015.597.
- [8] C. Delgado et al., Eds., *Book of Proceedings: 4th Edition of the IPS Pedagogical Practices Seminar*. Setúbal: Instituto Politécnico de Setúbal, 2026.
- [9] M. F. Menezes, “Proposal for a dashboard for analysis and visualization of educational data from Intelligent Tutoring Systems,” Master’s thesis, Institute of Computing, Federal University of Amazonas, Manaus, Brazil, 2024.